

# DESIGN AND MANUFACTURE OF MOBILE ACTUATOR FOR ANKLE EXOSKELETON

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**Abstract** - This research aims to design and manufacture a mobile actuator for an ankle exoskeleton and implement Analog control for the motor. Manufacturing techniques like sheet metal fabrication will be used to manufacture these housings. The actuator system will consist of 1 actuator for multi articular load paths of the ankle. The controller that will be used is the EPOS2 P controller from Maxxon. The actuator unit will comprise of EC 90 50/15 360W motor from Maxxon Motors. A planetary gearbox with a ratio of 50:1 will be used to get an output torque of 150Nm. The forces are transmitted using a Bowden cable. A battery will be mounted along with the motor in the same housing. These goals will be evaluated by carrying out step response test, maximum torque test. After designing the transmission system, the torque required for plantar flexion of exoskeleton will be calculated using a force sensor and the distance from the force sensor to pivot. Ankle angle will be measured using encoder and torque will be calculated using the above method. After designing the motor case and exosuit, subject will be tested for the metabolic cost to determine the performance of the suit.

**Keywords** – Motor Housing, Exoskeleton, Exosuit, CAD, Planetary Gearbox, Maxxon Motor

## I. INTRODUCTION

### A. Motivation

Over the past few years, development of exosuits have taken place at a rapid pace in helping the paralyzed people, person of disability and elders to walk and to reduce effort in lifting heavy objects. Major advancements in sensor technologies and actuator designs regarding their weight and size have enabled the commercialization of portable exosuits. The different types of exosuits that are available in the market are those that provide movement assistance to people who have lost movement completely, serving as physical therapy assistance units and that reduce fatigue and effort for normal humans to do repetitive tasks. In the first case patients have lost their legs completely our have been paralyzed below the hip so complete balance and safety are higher priority over speed. Considerable work has been conducted in the area of fully active exoskeletons for augmenting load carriage capacity. Key challenges for all these wearable exoskeletons are minimizing the weight and power requirement.[20]

This research aims to design and manufacture a mobile actuator for an ankle exoskeleton and implement Analog control for the motor. Manufacturing techniques like sheet metal fabrication will be used to manufacture these housings. The actuator system will consist of 1 actuator for multi articular load paths of the ankle. The controller that will be

used is the EPOS2 P controller from maxon. The actuator unit will comprise of EC 90 50/15 360W motor from Maxon Motors. A planetary gearbox with ratio of 50:1 will be used to get a output torque of 150Nm. The forces are transmitted using a Bowden cable. It works by the motor retracting the cable which reduces the distance between the two attached points creating assistance. A battery will be mounted along with the motor in the same housing. Load cells are mounted in series with the Bowden cable to measure the force delivered through the suit.[13]

### B. Literature Review

The Powered exoskeleton which is type of wearable robot which work in parallel with human muscles and combine the strength of robot with human Exosuits are the recent innovation which helps in human movement, by applying additional torques to the wearer through functional device. [4]. Over the last decade, there has been growing recognition of the importance of such gait assistive devices to increase the benefit to the wearer. The soft exosuit is made to help with plantarflexion, hip flexion, and hip extension while walking. A mobile actuation system is mounted on bag, and forces generated by the system are transmitted via cables to the exosuit. Many researchers are developing next generation soft wearable suits that use advanced technology and material to provide more comfortable, and effective mean to work together with human body.

Exosuits augments the capability of the healthy individuals. As compared to the older exosuits, these systems have several benefits [5]: can be align and adjusted easily, has low inertia, the joints of the wearer are unconstrained by the rigid structure, suit is very light. All these properties reduce the unintentional interference with the body's natural movement and allow for more effective interaction between the suit and human body. Wearable garments are important part of the suit as they provide means to transmit assistive torque to the joints. Force is transmitted over the body through optimum path such that biological-correct movement are produced at the joints. Some of the key features of the suits are actuated segments are extended, the length of the suit can increased do the entire suit is slack. It uses the shelf sensor technology like gyroscope, pressure sensor, IMU which detects the important movement in gait cycle [1] [5].



Fig 1: Motor Housing, Bowden Cable routing.

### C. Overview of the project

This research aims to improve to the design of the mobile actuator, manufacture a lightweight mobile actuation system and implement analog control. Our research aims to reduce the weight of the housing unit in which the actuator will be stored by using novel techniques like 3D printing to make the suit more comfortable and less bulky. Also, it aims at implementing analog control which will help to connect with high level controller. Previous researchers have made great strides in the design of the suit, but it is still very bulky and cumbersome to wear for long periods of time. We aim to overcome this in our research.

### D. Significance

**Software:** Analog input is required so that high-level controller can control the motor. The input from high-level controller such as sine wave is given to motor so motor can move in forward direction for positive sine wave and in reverse direction for negative sine wave.

**Motor Housing:** Our novel design of the motor housing reduces the weight and bulkiness of the mobile housing through FEM analysis. We use a compact transmission that make it comfortable to wear. Use of planetary gearbox and the high gear ratio it provides enables higher torque output.

**Harness:** The harness reduces the weight and bulkiness of the mobile housing using carbon fiber and M.O.L.L.E. plates. It can now support the weight of motor, its housing and gear and distribute their weight evenly on the body. Use of compact transmission makes it more comfortable to wear. Use of foam pads and vibration isolators makes it safe and avoid transmitting any forces or vibrations to the body. Also, the usage of padded straps make it lighter, comfortable as well as flexible.

The proposed project is based on further research and development in the field of Ankle foot Orthosis (AFO) by reducing the actuator weight and making it more usable, by this method we can reduce the metabolic compensation by 50% as less help from body is required to move. This kind of research will help the people with walking or locomotion. Especially there are millions of people who will be able locomotion. Furthermore, this can be an example of how simple design in robotics like this can assist humans using robotics. The assistive of this device helps humans to make movements easy and could be used by all people to lift heavy weight will long walks e.g. mountaineers, soldiers etc. The bio suit will be the prototype in research of providing the proper rehabilitation or permanent assistance depending on the need of the user.[18]

## II. METHOD

### A. Research Method

**Software:** The motor selected for Analog Control is EPOS2 P Maxon motor. The EPOS Studio on your computer communicates with these two devices by means of a USB2.0 serial connection. USB2.0 is the default communication. The code for Analog Control is written in Visual Studio in C++. Given in fig 2 is the Hardware Setup for Analog Control.

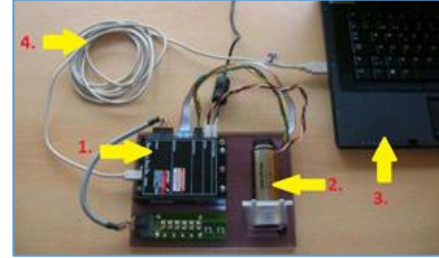


Fig 2: 1. EPOS2 P Controller 2. Maxon Motor Drive 3. Laptop 4. USB 2.0/3.0) [17]

In the code, initially communication is setup by communication handle, drive handle, gateway handle. Then the motor and analog velocity setpoint is enabled. Finally, analog control is activated. Given below is the pseudocode for analog control. [2]

Pseudocode: -[1]

1. def (Analog Control)
2. include "EposPCmd\_Comm.h"
3. include "EposPCmd\_Drive.h"
4. include <chrono>
5. include <string>
6. include <iostream>
7. # Initialization of header files
8. HANDLE CommunicationHandle
9. HANDLE DriveHandle
10. HANDLE GatewayHandle
11. DWORD ErrorCode
12. WORD NodeId
13. char GatewayProtocolStackName[]
14. DWORD Baudrate
15. DWORD Timeout
16. char ProtocolStackName[]
17. char InterfaceName[]
18. char PortName[]
19. char GatewayDeviceName[]
20. WORD MaxStrSize
21. char ErrorInfo;
22. # Data types
23. VCM\_OpenCommunication
24. VCM\_OpenGateway

25. VCM\_SetGatewaySettings
26. VCM\_OpenDrive
27. VCM\_SetDriveSettings
28. # Getting Correct Handles
29. VCM\_SetEnableState
30. VCM\_EnableAnalogVelocitySetpoint
31. # Enable Motor And Analog Velocity Setpoint
32. VCM\_ActivateVelocityMode
33. VCM\_ActivateAnalogVelocitySetpoint
34. # Activating Analog Control

After achieving analog control, the high-level controller (MATLAB/Simulink) is connected to EPOS2 P motor by using speedgoat cable by connections given in fig 3:

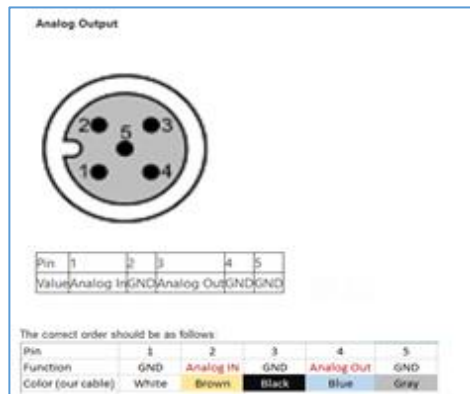


Fig 3: Speedgoat connections

So, using these connections, the blue and black wire is connected to ground and analog output of the EPOS2 P motor. So, the high-level controller is connected to low-level controller. Open loop analog control is achieved by using this process.

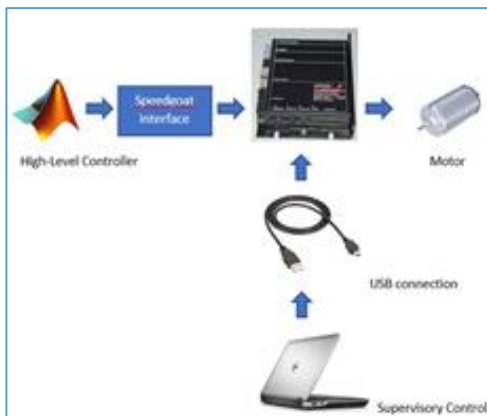


Fig 4: Open-loop analog control

To achieve closed loop analog control the ankle angle sensor and torque sensor needs to be connected back to high-level controller.

#### Motor Housing Design:

1) Iteration 1: We started our design process by designing motor housing and controller case. We designed the controller case as 3D printed structure. But to give more rigidity we decided to use aluminum plate.

2) Iteration 2: We designed the motor housing using aluminum plate and designed it for lighter weight by laser cutting the unnecessary portions. We used a single stage timing belt pulley system. The reason for using timing belt pulley system is its low cost.

3) Iteration 3: In this iteration we decided to use two stage pulley system since the single stage pulley system was becoming bulky.

4) Iteration 4: There was requirement of higher torque. Therefore, we needed to increase the stage in pulley system. We decided to go for 3 stage pulley system. We designed the pulley system as shown.

5) Iteration 5: The three-stage pulley system became bulky and out of dimensions. So we had to switch to a planetary gearbox for speed reduction. The positive point about the gearbox is that it is very compact. It can be attached to the harness perfectly.

TABLE I. Design Specifications

|                                     |                                                            |
|-------------------------------------|------------------------------------------------------------|
| Maximum Input motor torque          | 3 Nm                                                       |
| Gear ratio                          | 50:1                                                       |
| Obtained output torque              | $3 \times 50 = 150$ Nm maximum                             |
| Required output torque              | 140 Nm maximum                                             |
| Outer dimensions:                   | 168x89x99 mm                                               |
| Approximate weight of the assembly: | 2.9kg<br>(1.8(gearbox)+0.9(motor)+0.19 (other components)) |
| Efficiency of transmission:         | 94% [7]                                                    |
| Ventilation Fans                    | 40mm (Noctua NF-A4x10 FLXFan)                              |

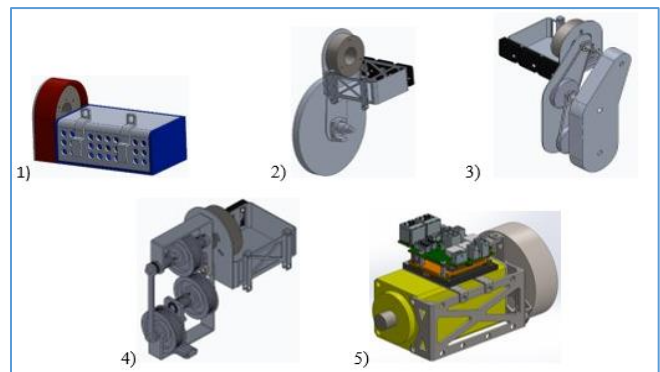


Fig 5: Design Iterations

**Harness Design:** In harness we started with straps attach in first phase as shown in figure given in fig 6. Here we were planned to attach the housing on the straps itself, but the distribution of weight was a big concern. Because of the complexity and no flexibility in design, with all forces and vibration transferred to hips we changes our design so that the weight is evenly distributed in the body. we found out M.O.L.L.E. which stands for modular load carrying system, here we solved the problem of weight distribution which is now distributed on shoulders and on the hip. The only problem we had with design now is that thickness of MOLLE plate was

not large enough to hold the weight of motor housing for a long period. We solved this problem by adding a Carbon fiber plate on a waist belt, which can hold the weight of the motor housing. We also had the option of using a Glass fiber plate but after comparing all the properties: weight, rigidity, toughness, and thermal characteristic we came to the conclusion that carbon fiber is a better choice. Its only drawback is that it's a little expensive.

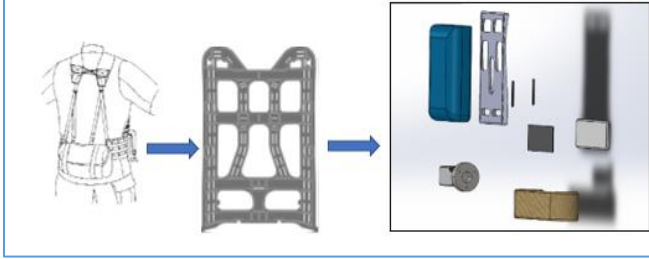


Fig 6: Harness Design

### B. Evaluation Method

After achieving analog control, performance characteristics of motor is calculated. The delay determines the amount of communication time required till motor is enabled. This is done by using Chronos timer and the delay can also be recorded by using EPOS studio software. The motor control is tested by sending a sine wave from a high-level controller to the low-level controller by using a speedgoat interface. Another method is converting C++ code into .mex file which is compatible with MATLAB/Simulink and the delay can be determined by using scope in Simulink. the methods of motor performance evaluation are the bench top test. We will carry out step response test, maximum torque test. After designing the transmission system, torque required for plantar flexion of exoskeleton will be calculated using force sensor and the distance from force sensor to pivot. Ankle angle will be measured using encoder and torque will be calculated using above method. There is also another method to calculate the torque. We can directly measure the torque using built-in feature of Maxon motor. We will compare the torque obtained by force sensor and the torque obtained by built in feature. If there is not much difference in those two values, then we will use the second method as it will reduce the sensing and computing time.

We will measure performance of the harness based on the weight of the harness, compact size, provision for attachments of accessories, durability of the fabric. [9]

After designing the motor case and exosuit, subject will be tested for metabolic cost to determine the performance of the suit. We will test for different conditions such as on tread mill, off tread mill, with backpack and without backpack. This can give us good understanding of metabolic cost.

## III. RESULTS

### A. Software

Two methods for delay result:

1. By using chronos timer, the delay is approx. 60 ms.
2. By using EPOS studio software, the results are given in fig 7:

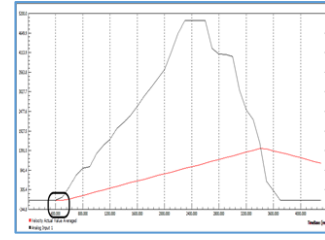


Fig 7: Graph between Time vs. velocity

TABLE II. Time vs Velocity data collected from motor

| Time     | Analog Input | Actual average velocity |
|----------|--------------|-------------------------|
| 0.0e+00  | 5.000000     | 6.000000                |
| 1.00e-01 | 6.000000     | 6.000000                |
| 2.00e-01 | 6.000000     | 6.000000                |
| 3.00e-01 | 7.000000     | 6.000000                |
| 4.00e-01 | 5.000000     | 6.000000                |
| 5.00e-01 | 89.000000    | 6.000000                |
| 6.00e-01 | 379.000000   | 43.000000               |
| 7.00e-01 | 700.000000   | 95.000000               |
| 8.00e-01 | 900.000000   | 141.000000              |

Black line denotes analog input and red line denotes actual average velocity. The main interest of the result is the black box on the graph. X-axis denotes the time and Y-axis denotes the velocity.

### B. Motor Housing

For the design of the housing CAD models were made using solidworks. The motor coupling, gearbox coupling, and back support was designed for minimum weight and maximum strength. Fig 8 shows the CAD models designed. The manufacturing process that will undertaken to make these models are laser cutting and bending. All the parts will be made from AISI 1020 steel which has a yield strength of 350MPa.

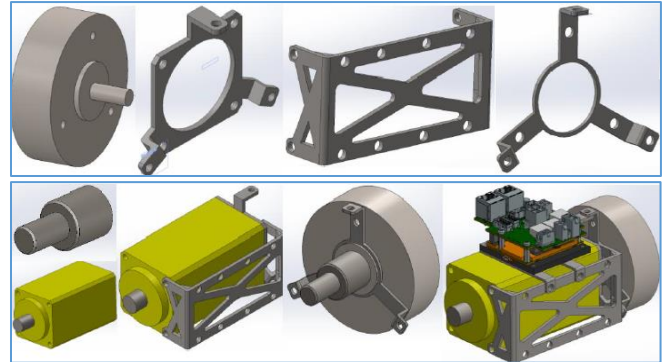


Fig 8: CAD Model

The assembly was iterated three times after the FEM analysis of the second iteration showed that the corners had a stress value of 390MPa which was higher than the yield strength shown in fig 9.

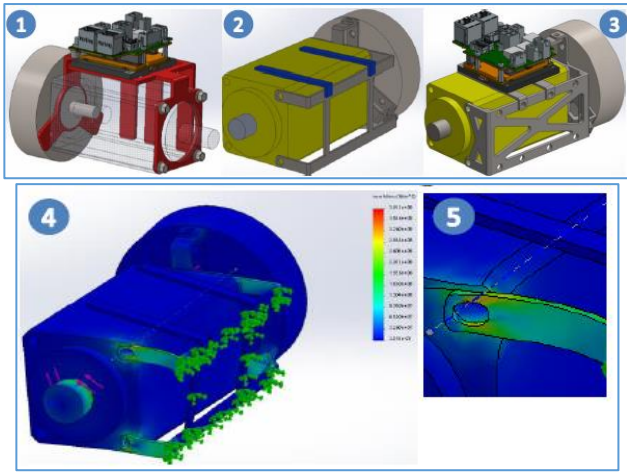


Fig 9: Design Iteration and FEM analysis of No.2

FEA was done on the final assembly to determine its strength under torque load of 150Nm at the output shaft and 3Nm torque at the motor shaft. From the results shown in fig 10 we can determine that the maximum stress on the assembly is 290MPa which is less than the 350MPa yield stress. Therefore, the design is safe.

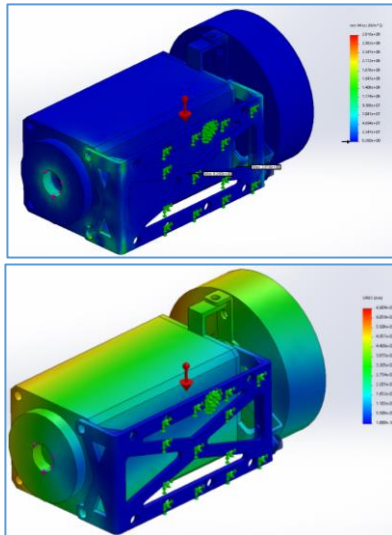


Fig 10: Stress Analysis, Displacement analysis.

### C. Harness Design

The harness design is as shown in fig 11. The 3 iterations of the design are shown, where the final design is combination of first two designs. Final design has MOLLE plate and carbon fiber plate on waist belt to support motor housing. Laptop bag is attached to the MOLLE plate. Route of Bowden cable is made in such a way that biological correct movement of the muscles is not restricted. Overall weight of suit is uniformly distributed over the body.

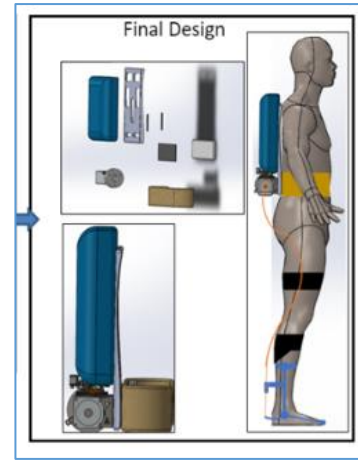
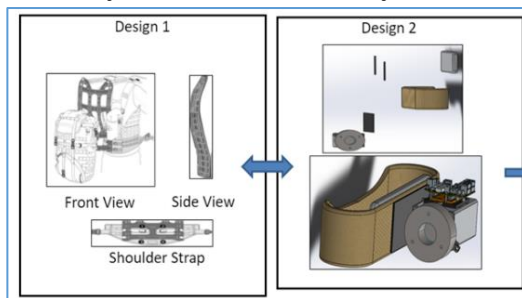


Fig 11: Harness Design Iterations

## IV. INTERPRETATION OF RESULTS

The reason for required voltage input is that, when we connect the low-level controller to a high-level controller the change in voltage from high level controller such as sine wave changes the speed accordingly in the low-level controller. The analog control is successfully achieved by this method. Then we find performance characteristics i.e. delay by using Chronos timer which is approx. 60 ms. The graph obtained from EPOS Studio software shows the delay. The interest is the black box it denotes the initialization of analog input and actual velocity averaged. From the table, the readings for 4.00e-01 and 5.00e-01 shows the initialization of the analog input and actual velocity average so the difference shows a delay of 100 ms. The results from EPOS studio are not accurate enough as it only provides a window of 170ms sampling period to record the data.

Our motor and gearbox housing design are very compact. It can be perfectly attached to the harness. It can be attached to the harness using nut bolts. It can sustain high applied torque. We performed FEA analysis and we observed that the stress due to torque of the motor is below the yield stress of the material. Therefore there won't be failure of the material. For harness and mobile exoskeleton design the design had many iterations, final design is quite flexible with safety to human body and economical. Also, the construction is easy, and fabrication of all parts is straight forward and will save cost and time. The structure has rigid next generation military grade weightlifting plate and carbon fiber plates to make it structurally perfect. The design still needs future improvements for adding safety when battery dies out or any part malfunction. Also, more Analysis could show some weak structure parts which could be improved.

## V. CONCLUSION/FUTURE DIRECTION

We designed a lightweight and strong mobile housing for the motor and transmission. FEA analysis proved the strength of the assembly. We performed analog control of the motor. We also designed the harness to be worn by the subject for mounting the housing on the body.

In real time event, the delay play as an important performance characteristics of motor control. The delay received by timer and EPOS studio software is approx. 60ms and 100ms respectively. But in real time this delay is considered very high. The solution for the problem is using EtherCAT which will considerably reduce delay upto 2 – 3 ms.

Mobile actuator can be designed for both legs. Therefore, there should be provision for actuator for another leg. Modular design for two motors can be done. There is also need of variable reduction ratio to check best suitable reduction ratio. The housing can be designed to accommodate different size gearbox. Weight of the system can be reduced by selecting light weight gearbox.

Force Analysis can be done on the harness. It could be made adjustable to fit different human sizes. The safety parameters can be increased to make the suit safer to wear for the subject.

## VI. LOGISTICS

### A. Member Contribution

Aniket: He was responsible for software designing and tuning of motor using C++ and epos studio. He performed data collection from analog control of motor. He was also responsible for interpreting results from the data. He also performed Finite Element Analysis of parts and did the 3D printing of the designed parts.

Abhishek: He was responsible for software designing and tuning of motor using C++ and epos studio. He was responsible for designing of motor housing, Design of Coupling and Mounts. He was a part of Idea Generation and Design Iteration.

Aditya Patki: Idea generation and Design Iterations. Designing of motor housing and transmission system. Finding the components according to design specifications for transmission. Finite Element Analysis of parts. Searching suitable Manufacturing facility.

Anirudh Joshi: literature survey, Designing and fabricating upper side and motor housing attachment, 3D printing Parts, Selection of Material for back support and Bowden Cable Routing.

Harshul Shah: Designing harness and study factors affecting its stability, selecting attachments, finding components to assemble flexible exosuit and fabrication of upper and Lower body side of exosuit. Also researching and studying many mobile exoskeletons, designs for harness and state of art designs.

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